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Department of Information Technology

COURSE CODE: DJS23ICPC501 **DATE: 31/07/2026**
COURSE NAME: Computer Networks **CLASS: T. Y. B.Tech**

Experiment No. 1

CO/LO: Demonstrate networking commands.

Aim: Study of Linux Networking Commands

EXERCISE

Try the following networking commands and discuss the use and importance of each.

1. hostname

   hostname

   aniket

2. nslookup

   nslookup

   google.com

   Server: 127.0.0.53

   Address: 127.0.0.53#53

   Non-authoritative answer:

   Name: google.com

   Address: 142.251.106.139

   Name: google.com

   Address: 142.251.106.100

   Name: google.com

   Address: 142.251.106.113

   Name: google.com

   Address: 142.251.106.102

   Name: google.com

   Address: 142.251.106.138

Name: google.com

Address: 142.251.106.101

Name: google.com

Address: 2404:6800:4000:1010::64

Name: google.com

Address: 2404:6800:4000:1010::66

Name: google.com

Address: 2404:6800:4000:1010::8a

Name: google.com

Address: 2404:6800:4000:1010::65

3. whois whois

Usage: whois [OPTION]... OBJECT...

-h HOST, --host HOST connect to server HOST
-p PORT, --port PORT connect to PORT
-I query whois.iana.org and follow its referral
-H hide legal disclaimers
--verbose explain what is being done
--no-recursion disable recursion from registry to registrar servers
--help display this help and exit
--version output version information and exit

These flags are supported by whois.ripe.net and some RIPE-like servers:
-l find the one level less specific match
-L find all levels less specific matches
-m find all one level more specific matches
-M find all levels of more specific matches
-c find the smallest match containing a mnt-irt attribute
-x exact match
-b return brief IP address ranges with abuse contact
-B turn off object filtering (show email addresses)
-G turn off grouping of associated objects
-d return DNS reverse delegation objects too
-i ATTR[,ATTR]... do an inverse look-up for specified ATTRibutes
-T TYPE[,TYPE]... only look for objects of TYPE

-K only primary keys are returned

-r turn off recursive look-ups for contact information -R

force to show local copy of the domain object even if

it contains referral

-a also search all the mirrored databases

-s SOURCE[,SOURCE]... search the database mirrored from SOURCE

-g SOURCE:FIRST-LAST find updates from SOURCE from serial FIRST to LAST

-t TYPE request template for object of TYPE

-v TYPE request verbose template for object of TYPE

-q [version|sources|types] query specified server info

4. route

   route

   Kernel IP routing table

   Destination Gateway Genmask Flags Metric Ref Use Iface

   default _gateway 0.0.0.0 UG 600 0 0 wlo1 link-local

   0.0.0.0 255.255.0.0 U 1000 0 0 wlo1

   172.17.0.0 0.0.0.0 255.255.0.0 U 0 0 0 docker0

   192.168.0.0 0.0.0.0 255.255.255.0 U 600 0 0 wlo1

5. arp

   arp

   Address HWtype HWaddress Flags Mask Iface

   _gateway ether 48:22:54:ee:00:18 C wlo1

6. ip ip

   addr

   1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

   inet 127.0.0.1/8 scope host lo valid_lft forever

   preferred_lft forever inet6 ::1/128 scope host

   valid_lft forever preferred_lft forever

   2: eno1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000

link/ether 24:6a:0e:df:95:84 brd ff:ff:ff:ff:ff:ff

altname enp4s0

3: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000

link/ether a8:59:5f:2a:5f:26 brd ff:ff:ff:ff:ff:ff

altname wlp0s20f3

inet 192.168.0.107/24 brd 192.168.0.255 scope global dynamic noprefixroute wlo1

valid_lft 5504sec preferred_lft 5504sec

inet6 fe80::1e49:c2b:f58f:3eb4/64 scope link noprefixroute

valid_lft forever preferred_lft forever

4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default

link/ether 36:ad:bf:17:45:f5 brd ff:ff:ff:ff:ff:ff

inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0

valid_lft forever preferred_lft forever

Conclusion : Study of network commands helps to understand the network configuration better

Reference:

[1] www.itsfoss.com